

Chapter 2 – Lab Requirements & Architecture

Before installing Wazuh, we must prepare a lab environment. A cybersecurity lab simulates a real enterprise network where different machines perform different roles.

Instead of installing everything on one computer, we create multiple virtual machines (VMs), each with a specific purpose.

2.1 Lab Requirements

Hardware Requirements (Recommended)

Component	Recommended
Processor	Intel/AMD with Virtualization (VT-x or AMD-V)
RAM	16 GB (Minimum 8 GB)
Storage	150 GB Free Space
Internet	Stable Connection
Virtualization Software	Oracle VirtualBox

2.2 Software Requirements

Download the following:

- Ubuntu Server 24.04 LTS
 - Windows 11 ISO
 - Kali Linux
 - VirtualBox
 - Wazuh Installation Packages (later)
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2.3 Why Do We Need Multiple Machines?

In a real company, one computer does not perform every task.

Different systems have different responsibilities.

For example:

- One server hosts websites.
- Another stores databases.
- Employees use Windows computers.
- Security teams monitor everything.
- Attackers attempt to compromise systems.

Our home lab follows the same idea.

2.4 Ubuntu Server

What is Ubuntu Server?

Ubuntu Server is a **Linux operating system** designed to host and manage services.

Unlike Ubuntu Desktop, it does not include a graphical desktop by default.

It is managed using the Linux terminal or SSH.

Purpose

Ubuntu Server is used to run services such as:

- Web Servers
- Database Servers
- Email Servers
- DNS Servers
- Docker
- Cloud Applications
- Wazuh Server

Why Do We Use Ubuntu Server?

Wazuh is designed to run as a server application.

Ubuntu Server provides:

- Stability
- Low resource usage
- Better performance
- Long-Term Support (LTS)
- Production-ready environment

In our lab, Ubuntu Server will host:

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard

Think of Ubuntu Server as the **brain of the SOC environment**.

2.5 Windows 11

What is Windows 11?

Windows 11 is Microsoft's desktop operating system used by millions of users worldwide.

Most organizations use Windows for employee workstations.

Purpose

Employees use Windows for:

- Office Work
- Browsing
- Email
- Business Applications

- Accessing Company Resources

Why Do We Use Windows?

Windows generates valuable security events such as:

- Login Attempts
- Failed Logins
- Event Logs
- PowerShell Activity
- Registry Changes
- Process Creation
- User Activity

These events are collected by the Wazuh Agent and sent to the Wazuh Manager for analysis.

Think of Windows as the **endpoint being monitored**.

2.6 Kali Linux

What is Kali Linux?

Kali Linux is a Linux distribution developed specifically for cybersecurity professionals.

Unlike Ubuntu Server, Kali includes a graphical interface and hundreds of pre-installed security tools.

Purpose

Kali is used for:

- Penetration Testing
- Vulnerability Assessment
- Digital Forensics

- Password Auditing
- Web Application Testing
- Network Scanning

Popular tools include:

- Nmap
- Metasploit
- Burp Suite
- Hydra
- Wireshark
- John the Ripper

Why Do We Use Kali?

Instead of waiting for real cyber attacks, we simulate attacks ourselves.

Examples:

- Port Scanning
- Brute Force Attacks
- Web Attacks
- File Modification
- Network Enumeration

These attacks generate security events that Wazuh detects.

Think of Kali as the **attacker machine (Red Team)**.

2.7 Windows Server (Optional but Important)

What is Windows Server?

Windows Server is Microsoft's server operating system.

Unlike Windows 11, it is designed to provide services to other computers on the network.

Purpose

Windows Server commonly hosts:

- Active Directory (AD)
- DNS
- DHCP
- File Sharing
- Group Policy
- Certificate Services

Why is Windows Server Important?

Most enterprise environments use Active Directory for identity and access management.

SOC analysts frequently investigate:

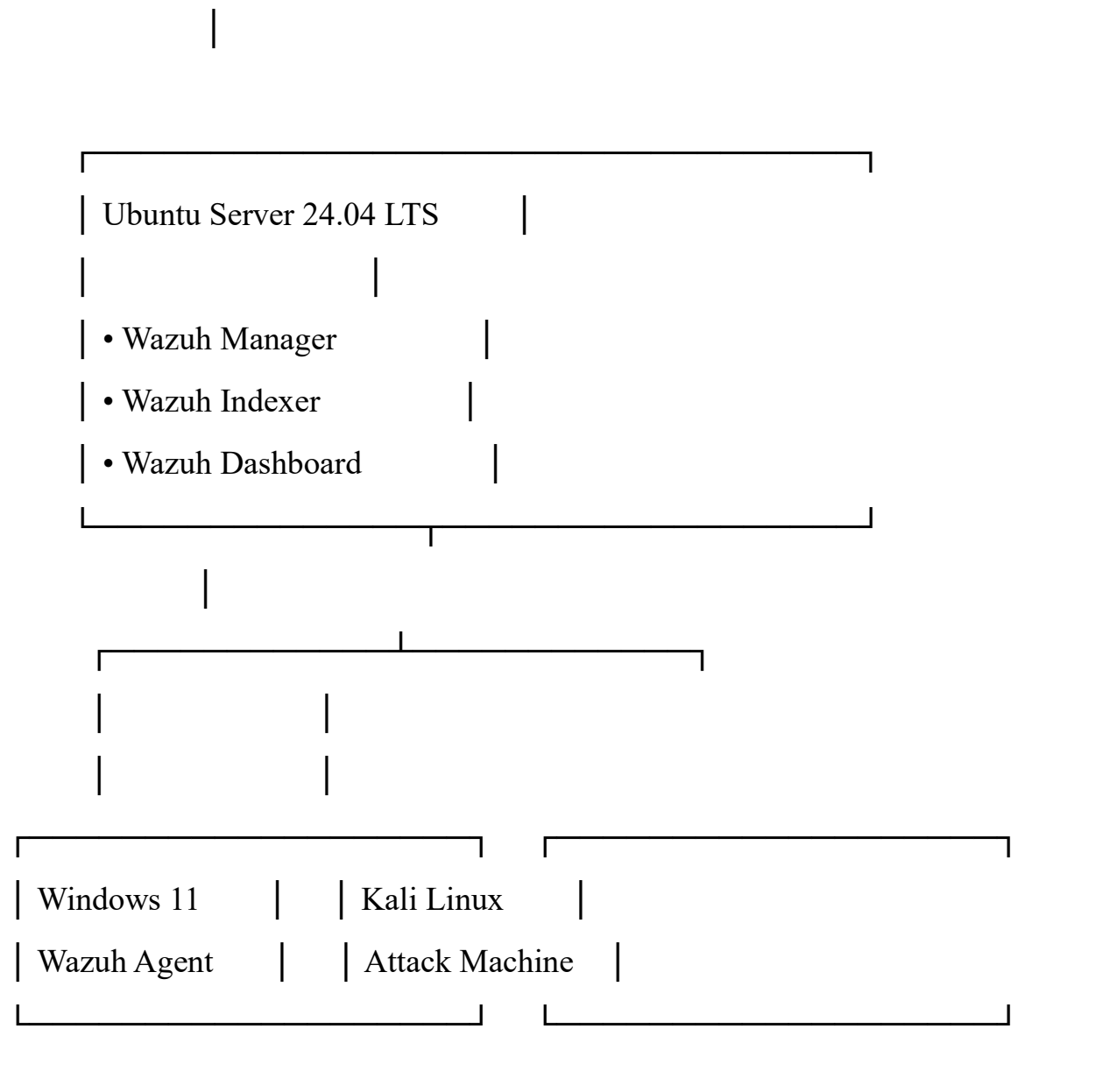
- Domain Logins
- Kerberos Authentication
- Group Policy Changes
- Domain Controller Events

Although we will not use Windows Server in the first version of our lab, we may add it later to simulate a real corporate network.

Think of Windows Server as the **identity and management server**.

2.8 Lab Architecture

VirtualBox



2.9 Role of Each Machine

Machine	Role
Ubuntu Server	Hosts Wazuh Manager, Indexer, and Dashboard
Windows 11	Generates logs and is monitored by Wazuh
Kali Linux	Simulates cyber attacks
Windows Server (Optional)	Provides Active Directory and enterprise services

2.10 How Data Flows

Kali Linux

(Attacker)→Windows 11(Victim)→Windows Event Log→Wazuh Agent→Ubuntu Server→(Wazuh Manager)Indexer→Dashboard→SOC Analyst

The SOC Analyst can investigate:

- Who attacked?
- Which IP Address?
- Which user was targeted?
- What files changed?
- What alerts were generated?

2.11 Why We Use Virtual Machines

Virtual Machines allow us to create multiple computers on one physical PC.

Benefits:

- Safe testing environment
- Easy snapshots and rollback
- Isolated from the host system
- No additional hardware required
- Similar to enterprise environments

Chapter 2 Summary

Before installing Wazuh, we prepare a realistic SOC lab.

The lab contains three main operating systems:

- **Ubuntu Server** → Hosts Wazuh services.

- **Windows 11** → Endpoint monitored by Wazuh.
- **Kali Linux** → Generates attacks for testing.

Later, **Windows Server** can be added to simulate enterprise services such as Active Directory.

Together, these systems create a complete environment where we can monitor, detect, investigate, and respond to cyber attacks.

One-Line Summary

A Wazuh home lab uses Ubuntu Server to host security services, Windows 11 as the monitored endpoint, Kali Linux to simulate attacks, and optionally Windows Server to provide enterprise services such as Active Directory.